Power BI Basics Assignment

Question 1 : What is Power BI and why is it used in businesses?

Ans - Power BI is Microsoft's business intelligence and analytics platform that helps businesses turn raw data into clear, actionable insights. It is widely used because it enables organizations to visualize data, monitor performance in real time, and make faster, fact-based decisions.

What Power BI Is

- Business Analytics Tool: Part of Microsoft's Power Platform, Power BI connects to multiple data sources (cloud, on-premises, Excel, SQL, etc.) and transforms them into interactive dashboards and reports.
- Visualization Engine: It converts complex datasets into charts, graphs, and maps that are easy to interpret.
- Integrated with Microsoft Ecosystem: Works seamlessly with Teams, Excel, SharePoint, and other Microsoft apps.

Why Businesses Use Power BI

- Data-Driven Decision Making
    - Provides real-time visibility into KPIs (sales, revenue, customer satisfaction).
    - Helps managers act quickly on trends or anomalies.
- Centralized Data Access
    - Combines data from multiple sources into a single "source of truth."
    - Eliminates silos and ensures consistency across departments.
- Ease of Use
    - Intuitive drag-and-drop interface suitable for both analysts and non-technical users.
    - Natural language queries allow users to ask questions like *"What were sales last quarter?"* and get instant answers.
- Scalability & Collaboration
    - Cloud-based service allows sharing dashboards across teams and devices.

- Supports enterprise-level governance, security, and compliance.
- AI-Powered Insights
  - Built-in Copilot features generate summaries, detect anomalies, and highlight reasons behind trends automatically.

Question 2 : Name and explain the three main components of Power BI.

Ans - The Three Main Components of Power BI

1. Power BI Desktop

- **What it is**: A free Windows application used to design and build reports.
- **Key Features**:
  - Connects to multiple data sources (Excel, SQL, cloud services).
  - Provides tools for data cleaning (Power Query), modeling (Power Pivot), and visualization (charts, maps, KPIs).
  - Allows offline report creation before publishing online.
- **Business Use**: Analysts and developers use it to prepare and design dashboards that can later be shared with teams.

2. Power BI Service (Cloud Platform)

- **What it is**: A cloud-based platform (app.powerbi.com) for publishing, sharing, and collaborating on reports.
- **Key Features**:
  - Enables creation of **dashboards** by pinning visuals from multiple reports.
  - Supports **real-time data refresh** and scheduled updates.
  - Provides **workspaces** for team collaboration and role-based access control.
  - Allows sharing via apps, links, or embedding in Microsoft Teams/SharePoint.
- **Business Use**: Managers and decision-makers use it to monitor KPIs, track performance, and collaborate securely across departments.

3. Power BI Mobile

- **What it is**: Mobile apps available for iOS, Android, and Windows devices.

- **Key Features**:
    - Access dashboards and reports on the go.
    - Interactive features like drill-down, filters, and alerts.
    - Push notifications for real-time updates.
- **Business Use**: Executives and field teams use it to stay updated on business metrics anytime, anywhere.

Question 3 : Explain the Power BI workflow.

Ans- Power BI Workflow Steps

1. Data Connection

- Connect to multiple data sources such as Excel, SQL Server, cloud services (Azure, Salesforce), or APIs.
- Power BI supports both on-premises and cloud-based data.
- This step ensures all relevant business data is accessible in one place.

2. Data Transformation & Modeling

- Use Power Query to clean, filter, and shape the data (remove duplicates, merge tables, handle missing values).
- Create relationships between tables using Power Pivot.
- Define calculated columns, measures, and hierarchies to make the data analysis-ready.

3. Data Visualization

- Build reports in Power BI Desktop using charts, graphs, maps, and KPIs.
- Create interactive dashboards by combining visuals.
- Apply slicers and filters to allow dynamic exploration of data.

4. Publishing & Sharing

- Publish reports from Power BI Desktop to the Power BI Service (cloud).
- Create dashboards by pinning visuals from different reports.
- Share dashboards with colleagues, embed them in Teams/SharePoint, or set up scheduled refreshes.

5. Access Anywhere

- Use Power BI Mobile to view dashboards on phones and tablets.
- Receive alerts and notifications for KPI changes in real time.

Question 4 : List any four data cleaning tasks that can be performed in Power Query.

Ans- Four Data Cleaning Tasks in Power Query

1. Remove Duplicates

- Eliminates repeated rows or values from a dataset.
- Example: If customer records appear multiple times, you can keep only the unique entries.

2. Handle Missing Values

- Replace nulls with default values, blanks, or calculated substitutes.
- Example: Fill missing sales figures with "0" or use the average of existing values.

3. Change Data Types

- Convert columns into the correct format (text, number, date, currency).
- Example: Transform a "Date" column stored as text into an actual date type for proper sorting and filtering.

4. Split or Merge Columns

- Break down a single column into multiple parts or combine several columns into one.
- Example: Split a "Full Name" column into "First Name" and "Last Name," or merge "City" and "State" into one "Location" column.

Question 5 : Write step by step instructions to load the above dataset.

Ans- To load the dataset follow the below steps-

1. Open the Power BI desktop.
2. Choose get data from ribbon.
3. Select text/csv.
4. Choose load data.

Question 6 : Define Data View, Report View, and Model View. Explain the purpose of each view.

Ans- 1. Data View

- Definition: Displays the actual data tables you've loaded into Power BI.
- Purpose:
  - Lets you inspect rows and columns directly.
  - Useful for verifying data after cleaning and transformation.
  - Allows creation of calculated columns and measures using DAX (Data Analysis Expressions).
- Example Use: Checking if missing values were replaced correctly or if a calculated "Profit" column is working as expected.

2. Report View

- Definition: The canvas where you design and build interactive reports and dashboards.
- Purpose:
  - Drag and drop fields to create visuals (charts, maps, KPIs).
  - Apply filters, slicers, and formatting.
  - Arrange visuals into dashboards for storytelling.
- Example Use: Creating a sales dashboard with regional charts, product slicers, and monthly trend lines.

3. Model View

- Definition: Shows the relationships between different tables in your dataset.
- Purpose:
  - Manage relationships (one-to-many, many-to-many).
  - Define hierarchies and set cross-filtering rules.
  - Ensures data integrity across multiple tables.
- Example Use: Linking a "Sales" table with a "Products" table through the "ProductID" field so visuals can combine both datasets correctly.

Question 7 : Discuss the different data sources that Power BI supports.

Ans- Power BI supports a wide range of data sources including files (Excel, CSV, JSON), databases (SQL Server, Oracle, MySQL, PostgreSQL), cloud services (Azure, Salesforce, Google BigQuery), and online platforms (SharePoint, Dynamics 365, GitHub, etc.). This flexibility allows businesses to integrate data from virtually any system into one unified analytics environment.

Categories of Data Sources in Power BI

1. File-Based Sources

- Excel Workbook
- Text/CSV
- XML, JSON
- PDF, Parquet
- SharePoint Folder Useful for importing structured or semi-structured data stored locally or in cloud storage.

2. Database Sources

- SQL Server, SQL Server Analysis Services
- Oracle Database, IBM Db2, IBM Netezza
- MySQL, PostgreSQL, Sybase, Teradata
- SAP HANA, SAP Business Warehouse
- Amazon Redshift, Google BigQuery, Snowflake, Vertica Ideal for enterprise-level structured data stored in relational or analytical databases.

3. Microsoft Fabric & Power Platform

- Dataverse (formerly Common Data Service)
- Power BI datasets
- Power Platform connectors (Power Apps, Power Automate) Enables integration with Microsoft's ecosystem for seamless workflow automation.

4. Azure Cloud Services

- Azure SQL Database, Azure Synapse Analytics
- Azure Blob Storage, Azure Data Lake, Azure Cosmos DB
- Azure Analysis Services, Azure HDInsight Provides scalable cloud-based data storage and analytics integration.

5. Online Services

- Dynamics 365 (CRM, ERP)
- Salesforce Objects & Reports
- Google Analytics, Facebook, GitHub, MailChimp
- Smartsheet, Projectplace, Mixpanel Useful for connecting to SaaS platforms and third-party applications.

6. Other Sources

- Web data (scraping tables from websites)
- OData feeds
- ODBC/OLE DB connectors
- R and Python scripts Extends Power BI's capabilities to custom or advanced data sources.

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

Ans- To split the Owner Name column into First Name and Last Name in Power BI using Power Query, follow these steps:

Step-by-Step Instructions

1. Load Your Dataset

- Open Power BI Desktop.
- Load your dataset into Power BI (via Excel, CSV, SQL, etc.).
- Go to Home → Transform Data to open Power Query Editor.

2. Select the Column

- In Power Query, locate the Owner Name column.
- This column contains full names (e.g., *John Smith*).

3. Use Split Column Feature

- Right-click on Owner Name → choose Split Column.
- Select By Delimiter.
- Choose Space as the delimiter (since names are separated by spaces).

4. Configure Split

- Choose Split at the left-most delimiter (for First Name).
- Choose Split at the right-most delimiter (for Last Name).
- Power Query will create two new columns: Owner Name.1 and Owner Name.2.

5. Rename Columns

- Rename Owner Name.1 → First Name.
- Rename Owner Name.2 → Last Name.

6. Apply Changes

- Click Close & Apply to load the transformed data back into Power BI.
- You now have two separate columns for First Name and Last Name.